

Project 2: Bioinformatic Analysis

Sample ID: S17

1. Introduction

In this project, we analyzed DNA sequencing data from a cancer patient. We had two samples: one from healthy tissue and one from the tumor. Our goal was to find mutations that exist only in the tumor, these are called somatic mutations and to understand which of them might be causing the cancer.

2. Input Data

File	Description	Reads	Total Bases
S17.C_R1.fastq.gz	Normal tissue, Read 1	46,678,977	4.7 Gbp
S17.C_R2.fastq.gz	Normal tissue, Read 2	46,678,977	4.7 Gbp
S17.T_R1.fastq.gz	Tumor tissue, Read 1	~93,500,000	~9.4 Gbp
S17.T_R2.fastq.gz	Tumor tissue, Read 2	~93,500,000	~9.4 Gbp

All samples were paired-end Illumina sequencing data (Sanger / Illumina 1.9 encoding). The higher read count in tumor compared to normal is consistent with deeper sequencing coverage typical in cancer samples.

3. Analysis Pipeline

The following pipeline was implemented following GATK Best Practices for Somatic Short Variant Discovery:

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FASTQ → FastQC → BWA MEM → SAMtools sort → GATK MarkDuplicates → GATK Mutect2 →
FilterMutectCalls → VEP Annotation
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Tool	Version	Purpose
FastQC	0.11.x	Raw data quality control
BWA MEM	0.7.19	Read alignment to hg38
SAMtools	1.x	BAM sorting and indexing

GATK MarkDuplicates	4.6.2.0	PCR duplicate marking
GATK Mutect2	4.6.2.0	Somatic variant calling (tumor vs normal)
GATK FilterMutectCalls	4.6.2.0	Variant filtration
Ensembl VEP	v115	Variant annotation (COSMIC 101, ClinVar 202502)

Reference genome: GRCh38/hg38

4. Raw Data Quality Control

4.1 FastQC Summary

FastQC was run on all four FASTQ files. Results were consistent across Normal R1/R2 and Tumor R1/R2:

Module	Normal R1	Normal R2	Tumor R1	Tumor R2
Basic Statistics	PASS	PASS	PASS	PASS
Per base sequence quality	PASS	PASS	PASS	PASS
Per tile sequence quality	PASS	PASS	PASS	PASS
Per sequence quality scores	PASS	PASS	PASS	PASS
Per base sequence content	PASS	PASS	PASS	PASS
Per sequence GC content	FAIL	FAIL	FAIL	FAIL
Per base N content	PASS	PASS	PASS	PASS
Sequence Length Distribution	WARN	WARN	WARN	WARN
Sequence Duplication Levels	PASS	PASS	PASS	PASS
Overrepresented sequences	PASS	PASS	PASS	PASS
Adapter Content	PASS	PASS	PASS	PASS

4.2 Basic Statistics (Normal Sample R1)

Metric	Value
Total sequences	46,678,977
Total bases	4.7 Gbp

Sequences flagged as poor quality	0
Sequence length	18–101 bp
%GC	49%

4.3 Interpretation of QC Results

Per base sequence quality (PASS): Sequence quality scores were consistently high across all read positions in all four samples, indicating high-quality sequencing data. No quality trimming was required, and reads were used directly for alignment.

Adapter Content (PASS): No adapter contamination was detected in any of the samples. Trimming with Trimmomatic was not performed.

Sequence Length Distribution (WARN): Read lengths ranged from 18 to 101 bp rather than being uniformly 101 bp. This is a common observation in real-world sequencing data where some reads are trimmed during base calling. It does not negatively affect downstream analysis when read lengths are sufficient for alignment (>30 bp for most reads).

Per sequence GC content (FAIL): The GC content distribution deviated from the expected theoretical distribution in all four samples, including the matched normal. This systematic deviation, present in both tumor and normal, suggests it is likely a technical characteristic of the sequencing library preparation or the specific genomic regions captured by the exome kit, rather than a biological artifact specific to the tumor. Notably, the human exome is enriched for GC-rich regions, which can shift the observed GC distribution. This result does not compromise the quality of the variant calling analysis.

5. Read Alignment

We aligned the sequencing reads to the human reference genome (hg38) using a tool called BWA MEM. This step maps each short DNA read to its correct position in the genome. Both samples aligned successfully, the normal sample produced a 7.1 GB file and the tumor a 14 GB file, reflecting the higher sequencing depth of the tumor. The reads in both samples showed the expected paired-end orientation, confirming that the library preparation was correct.

Metric	Normal	Tumor
Sorted BAM size	7.1 GB	14 GB
Insert size (mean)	162.77 bp	N/A
Insert size (std dev)	45.44 bp	N/A
Read orientation	FR (standard Illumina)	FR (standard Illumina)

6. Post-processing

6.1 PCR Duplicate Marking

GATK MarkDuplicates was used to identify and flag PCR duplicates. Duplicates are marked but not removed, allowing downstream tools to appropriately exclude them from variant calling calculations.

Metric	Normal	Tumor
Total reads processed	93,233,386	185,763,619
Marked duplicates	6,209,561	52,415,650
Duplication rate	6.7%	28.2%
Optical duplicates	4,702	2,434,035

Normal sample (6.7% duplication): This is well within the acceptable range and indicates good library complexity. Low duplication rates ensure that variant allele frequencies accurately reflect the true biology.

Tumor sample (28.2% duplication): The substantially higher duplication rate in the tumor sample, along with a dramatically higher optical duplicate count (2,434,035 vs 4,702 in normal), suggests two possible explanations: (1) the tumor sample was sequenced at approximately 2x greater depth than normal, increasing the probability of PCR duplicates, and (2) potential copy number amplifications in tumor regions could result in regions being over-sequenced and flagged as duplicates. Despite the higher duplication rate, 71.8% of reads remained unique and available for variant calling.

7. Post-Alignment Quality Control

7.1 Flagstat Results

Metric	Normal	Tumor
Total reads	93,398,230	187,138,941
Mapped reads	93,214,273 (99.80%)	185,611,820 (99.18%)

Properly paired	92,834,026 (99.40%)	184,850,495 (98.78%)
Singletons	19,403 (0.02%)	152,820 (0.08%)
Mate mapped to different chr	131,427	374,079

Both samples demonstrated excellent alignment metrics. Mapped read rates exceeding 99% in both samples indicate successful alignment to the reference genome hg38. The properly paired rates above 98% confirm that the paired-end sequencing data is of high quality. The low singleton rates (<0.1%) indicate minimal read loss during alignment. The small proportion of reads mapping to different chromosomes (<0.2%) is within normal range and does not affect downstream analysis.

8. Variant Calling

8.1 Mutect2 Somatic Variant Calling

Somatic variants were identified using GATK Mutect2 in tumor-normal paired mode. This approach compares tumor BAM against matched normal BAM to identify somatic mutations absent from the germline, minimizing false positives from germline variants.

Step	Variants
Raw Mutect2 output	23,412
After FilterMutectCalls (PASS)	572
Filter rate	97.6%

The high filter rate (97.6%) is expected in whole-exome sequencing without a Panel of Normals (PON) or germline resource, as Mutect2 applies conservative statistical filters to minimize false positives. The 572 PASS variants represent high-confidence somatic mutations for downstream annotation and interpretation.

9. Variant Annotation and Interpretation

9.1 Annotation with Ensembl VEP

Filtered variants (somatic.filtered.vcf.gz) were annotated using Ensembl VEP v115 against GRCh38 with the following databases: COSMIC v101 (known cancer somatic mutations), ClinVar (202502), dbSNP 156, gnomAD v4.1, PolyPhen-2, and SIFT. VEP predicted functional consequences and assigned impact levels (HIGH, MODERATE, LOW, MODIFIER) to each variant.

9.2 Summary of Annotated Variants by Impact

Impact Level	Count	Description
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HIGH	~150 unique	Stop gains, frameshifts, splice site disruptions
MODERATE	~400 unique	Missense variants, inframe indels
LOW	Majority	Synonymous variants, splice region variants
MODIFIER	Majority	Intronic, upstream, downstream variants

9.3 Cancer-Relevant Mutations

The following high-priority variants were identified in known cancer-associated genes:

Gene	Consequence	Impact	ClinVar	Function
KMT2C	frameshift_variant	HIGH	pathogenic	Histone H3K4 methyltransferase, chromatin remodeling
KMT2C	stop_gained	HIGH	-	Premature termination of KMT2C protein
KMT2C	splice_acceptor_variant	HIGH	uncertain_significance	Disruption of mRNA splicing
ARID1A	inframe_deletion	MODERATE	-	SWI/SNF chromatin remodeling complex
ARID1A	missense_variant	MODERATE	-	SIFT: deleterious, PolyPhen: possibly_damaging
BCOR	stop_gained	HIGH	-	Transcriptional corepressor, X-linked tumor suppressor
MKI67	frameshift_variant	HIGH	-	Cell proliferation marker (Ki-67)
CIC	frameshift_variant	HIGH	-	Transcriptional repressor, tumor suppressor

9.4 Biological Interpretation

Based on the annotation results, the most important mutations affect genes that control how DNA is organized and which genes are active in the cell.

KMT2C: We found three different mutations in this gene, a frameshift, a stop codon, and a splice site mutation. All three are classified as pathogenic in ClinVar. KMT2C normally adds chemical marks to DNA packaging proteins (histones) to activate nearby genes. When KMT2C is lost, large parts of the genome lose their normal activity control.

ARID1A: This gene is part of a protein complex called SWI/SNF, which physically moves and repositions DNA packaging to allow genes to be read. It is one of the most commonly

mutated tumor suppressor genes in human cancer. We found a deletion and a missense mutation, both predicted to damage the protein.

BCOR and CIC: Both of these genes act as brakes on gene expression. BCOR has a premature stop codon and CIC has a frameshift mutation, in both cases the protein is likely completely destroyed through a process called nonsense-mediated decay.

Overall, the tumor shows a pattern where multiple genes responsible for controlling the epigenetic state of the cell are simultaneously damaged. This kind of widespread epigenetic failure is characteristic of endometrial carcinoma, colorectal cancer, or ovarian clear cell carcinoma.

10. Limitations

- Structural variant (SV) analysis and copy number variation (CNV) calling were not performed due to time constraints. Tools such as Manta (SV) and CNVkit (CNV) would be required for a complete analysis.
- No Panel of Normals (PON) was used in Mutect2, which may result in a higher false positive rate for technical artifacts.
- Base Quality Score Recalibration (BQSR) was not performed due to the absence of a known variants VCF file, which may slightly affect variant quality scores.

11. Conclusion

In this project, we ran a complete analysis pipeline on sequencing data from a cancer patient. Both samples were of high quality, with over 99% of reads successfully mapped to the reference genome. The tumor sample had a higher duplicate rate (28.2% vs 6.7%), likely due to deeper sequencing or copy number changes in the tumor.

After variant calling and filtering, we identified 572 high-confidence somatic mutations. The most important findings were mutations in KMT2C, ARID1A, BCOR, and CIC, genes that normally control how DNA is packaged and which genes are turned on or off. When these genes are damaged, cells can lose control over their growth and become cancerous.

These findings suggest the patient may benefit from immunotherapy or drugs that target epigenetic pathways, such as EZH2 inhibitors.

12. References

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